5.2 Towards Useful and Identifiable Interpretability Tasks

Identifiability is not an intrinsic property of the model under study but of the interaction between μ, $\mathcal{E}, D$, the model f_{NN}, and the behavior of interest $P_{\mathbf{U}}$. We might wonder what choices to make in order to improve the identifiability and usefulness of interpretability queries.

Query richness. The queries in the support of μ must be sufficiently discriminative to distinguish candidate explanations in $\mathcal{E}$. There is a fundamental trade-off between discriminative power and sample efficiency. If μ spreads probability mass over a large support, accurately estimating L_μ may require prohibitive amounts of interventional data. Conversely, concentrating μ on too few queries risks not singling out one explanation in $\mathcal{E}$.

Expressivity vs. parsimony in $\mathcal{E}$. Conversely, the hypothesis class must have sufficient capacity to approximate the queries well (low bias) but not so much flexibility that many distinct explanations all achieve low error (large equivalence classes, high variance, non-identifiability). This is akin to the classical bias-variance tradeoff, pointing toward standard fixes like regularization of the hypothesis class (Belinkov, 2022).

Human cognitive constraints. Interpretability is meant to facilitate *human understanding*. Empirical studies suggest people can mentally simulate models with only a handful of interacting components (Lombrozo, 2006; Wilkenfeld, 2013; Keil, 2006; Hassija et al., 2024). Explanations exceeding these structural limits may be technically *correct* yet fail to provide insight. Designing $\mathcal{E}$ with human simulability in mind ensures that understanding remains the end goal.

5.3 Opportunities for Future Work

Characterizing identifiability conditions. A systematic theoretical program could characterize when specific $(\mu, \mathcal{E}, D)$ triplets are identifiable, mirroring similar efforts in causal inference (Shpitser & Pearl, 2008) and unsupervised learning (Locatello et al., 2019; Khemakhem et al., 2020). What symmetries and invariances are unavoidable in representation space, and when is identifiability up to such equivalences acceptable? Constructing a taxonomy of identifiable interpretability tasks would provide actionable guidance for practical scenarios.

Bayesian interpretability and uncertainty quantification. Bayesian approaches offer an elegant framework for handling non-identifiability and quantifying uncertainty (Gelman et al., 2013). Specifically, one could specify a **prior distribution** $\pi(e)$ over the explanation class $\mathcal{E}$, encoding structural preferences (e.g., sparsity, modularity) or incorporating prior information from related studies. Then, the **likelihood model** $P(\mathcal{T}_n \mid e)$ describes how computational traces are generated given explanation e. Finally, the **posterior updates** via Bayes' rule: $\pi(e \mid \mathcal{T}_n) \propto P(\mathcal{T}_n \mid e)\pi(e)$, refines beliefs as observations accumulate. Then, **credible sets** can quantify uncertainty. When explanations are non-identifiable, the posterior remains diffuse across an equivalence class; uncertainty quantification naturally reflects this fundamental ambiguity. Conversely, as more discriminative queries are observed, the posterior concentrates. This further provides a principled framework for active setup: strategically selecting queries from μ that maximally reduce posterior uncertainty.

Meta-analysis and cumulative science. Meta-analytic methods (Borenstein et al., 2021) could coherently aggregate evidence across studies, accounting for heterogeneity in μ, $\mathcal{E}$, and experimental conditions. Standardized effect size measures, pre-registration of analyses, and open sharing of collected computational traces would enable interpretability to become a cumulative science where knowledge systematically builds over time. In general, the solutions proposed by other fields (Poldrack et al., 2017; Korbmacher et al., 2023) discussed in the introduction now become available for interpretability.

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